

SARMASU, Romania

WMO: 151230

Lat: 46.7475N

Lon: 24.1597E

Elev: 400

StdP: 96.61

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.4	-15.9	1.0	-11.9	-13.1	1.3	-9.3	9.2	2.9	7.7	0.3	1.0	320	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	31.4	20.3	29.9	19.9	28.2	19.2	21.7	29.0	20.9	27.9	20.1	26.7	2.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.8	24.8	18.6	14.1	23.6	17.9	13.5	22.7	65.1	29.3	62.1	28.0	59.4	26.7	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	6.9	5.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-2.4	1.1	5.4	10.7	15.9	19.4	20.3	21.2	16.0	10.5	5.4	-0.3		
	DBStd	9.00	4.47	4.62	4.31	5.01	3.45	3.16	2.90	3.27	4.44	3.90	4.68	4.07		
	HDD10.0	1354	385	248	150	53	3	0	0	0	3	43	151	319		
	HDD18.3	3203	643	482	401	232	89	24	15	12	97	243	389	577		
	CDD10.0	1468	0	1	8	72	187	281	319	346	184	58	12	0		
	CDD18.3	274	0	0	0	1	14	56	76	99	28	0	0	0		
	CDH23.3	2367	0	0	0	27	128	464	584	931	229	3	0	0		
	CDH26.7	728	0	0	0	2	18	127	161	360	60	0	0	0		
(14) Wind	WSAvg	2.0	1.7	2.1	2.4	2.8	2.1	1.9	1.8	1.8	1.9	1.8	1.8	1.9		
(15) Precipitation	PrecAvg	561	24	20	28	45	63	86	81	47	53	44	34	31		
	PrecMax	749	51	48	70	79	111	195	122	93	128	91	65	65		
	PrecMin	385	7	5	5	3	18	16	24	7	8	1	0	4		
	PrecStd	99	13	11	17	20	23	36	32	25	29	24	16	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	14.2	20.0	26.4	28.6	31.7	32.3	34.1	31.1	23.5	18.2	9.9		
		MCWB	6.3	9.5	11.7	15.0	16.5	21.7	21.6	21.2	18.6	15.3	12.6	7.9		
	2%	DB	6.1	11.4	16.8	23.5	26.5	29.9	30.2	32.1	28.2	21.6	15.7	7.8		
		MCWB	4.6	7.8	9.5	14.2	17.0	20.6	20.8	19.9	18.0	14.5	11.4	6.3		
	5%	DB	4.5	9.3	14.0	21.1	24.6	28.1	28.4	30.4	25.8	19.6	13.9	6.1		
		MCWB	3.3	6.6	8.3	13.2	16.2	19.5	19.9	19.7	16.8	13.7	10.4	4.8		
	10%	DB	2.9	7.4	11.7	18.9	22.6	26.2	26.6	28.5	23.4	17.2	12.0	4.4		
		MCWB	1.9	5.3	6.9	11.9	15.4	18.8	19.2	19.4	16.3	12.5	9.3	3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	10.2	12.3	16.1	18.9	23.4	22.7	23.0	20.5	16.6	13.1	8.2		
		MCDB	7.9	13.7	18.8	23.5	24.7	29.8	30.3	30.8	29.4	21.2	16.8	9.4		
	2%	WB	4.8	8.0	10.2	15.0	17.7	21.5	21.5	21.5	18.9	15.3	12.0	6.3		
		MCDB	5.9	10.9	15.7	22.3	24.4	28.1	28.4	29.7	25.0	20.0	15.3	7.6		
	5%	WB	3.4	6.7	8.8	14.0	16.9	20.4	20.7	20.7	17.8	14.2	10.9	4.9		
		MCDB	4.3	8.9	13.2	20.1	23.1	26.8	27.0	28.5	23.8	19.0	13.6	5.9		
	10%	WB	2.0	5.4	7.6	12.6	16.0	19.4	20.0	19.8	16.9	13.0	9.6	3.7		
		MCDB	2.7	7.2	11.0	17.8	21.2	25.3	25.3	27.0	22.3	16.6	12.0	4.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.8	9.3	10.6	10.2	10.7	11.1	12.0	10.1	9.5	6.5	4.3		
		MCDBR	5.8	9.4	13.1	14.3	13.1	13.5	13.5	14.9	13.6	13.8	9.2	6.3		
	5% WB	MCWBR	4.5	6.2	7.2	6.7	5.4	5.6	5.2	5.0	5.0	7.1	5.6	4.9		
		MCWBR	5.7	8.6	11.6	13.4	12.0	12.5	12.3	13.8	12.7	12.5	8.1	6.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.309	0.335	0.375	0.420	0.405	0.402	0.421	0.425	0.387	0.359	0.336	0.311		
		taud	2.336	2.300	2.255	2.166	2.268	2.305	2.255	2.243	2.319	2.384	2.396	2.367		
	Edn at Noon	Edn at Noon	772	824	841	833	861	864	840	820	819	789	735	725		
		Edh at Noon	75	95	116	139	130	126	131	128	108	87	71	66		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.23	2.07	3.19	4.33	5.31	5.77	5.70	5.25	3.73	2.54	1.43	0.94		
	RadStd	RadStd	0.17	0.24	0.41	0.43	0.50	0.47	0.50	0.48	0.43	0.32	0.19	0.12		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (66 neighbors)		+0.75	N/A	N/A	+0.97	N/A	N/A	-157	-228	+92	+18	

Nomenclature: See separate page